

PAVAN KALYAN NEELAM

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[LinkedIn](#) | [GitHub](#)

SUMMARY

AI Engineer with hands-on experience building Generative AI, Voice AI, RAG pipelines, and AI agents. Skilled in developing production-grade conversational AI systems, optimizing retrieval architectures, and deploying scalable LLM-powered applications and workflows.

TECHNICAL SKILLS

Programming: Python, SQL

AI / ML & NLP: PyTorch, scikit-learn, Transformers, LangChain, RAG, MCP, Prompt Engineering, OpenAI API, Guardrails, FT

Voice AI: VAPI, LiveKit, Speech-to-Text (STT), Text-to-Speech (TTS)

Data Analysis: NumPy, Pandas, Matplotlib, EDA, Data Cleaning & Transformation

Databases & Vector Stores: MySQL, PostgreSQL, Qdrant, ChromaDB

Tools & Platforms: Jupyter Notebook, VS Code, Git, GitHub Actions

Soft Skills: Communication, Problem-Solving, Teamwork, Time Management, Analytical Thinking

EXPERIENCE

Seanergy.AI — AI/ML Intern | Hyderabad, India

Feb 2026 – Present

MYURI – LiveKit Voice Agent Platform | LiveKit, RAG, MCP, LLMs, STT/TTS

Apr 2026 – Present

- Optimized RAG pipelines to improve retrieval performance for voice-agent applications.
- Implemented PgBouncer connection pooling for efficient PostgreSQL database management.
- Replaced LiveKit Egress with RecorderIO to streamline conversation-recording workflows.

MedConcepts – Healthcare Voice Agents | Vapi, OpenAI API

Apr 2026 – May 2026

- Built healthcare voice agents for surgery scheduling and pre-operative health screening using Vapi and LLM-based conversational workflows.
- Implemented patient verification, clinical response capture, tool calling, escalation handling, and automated status management.

Infosys Springboard — AI Intern | Remote

Aug 2025 – Oct 2025

Emotion Detection & Music Recommendation System | Python, CNN, Gemini API

- Built an emotion detection system using CNNs to analyze facial expressions and mood, recommending music accordingly.
- Implemented data preprocessing, model training, and recommendation logic for an end-to-end AI-driven solution.

Prasunet Pvt. Ltd. — Data Science Intern | Chandigarh

June 2024 – July 2024

- Developed and evaluated ML models in Python (scikit-learn), achieving a 10% improvement in predictive accuracy.
- Performed EDA, preprocessing, and visualization using Pandas and Matplotlib, reducing data-cleaning time by 15%.

PROJECTS

Conversational AI Interview – Sarah AI | Python, FastAPI, OpenAI API

Sep 2025

Built an AI-driven virtual interviewer using GPT and FastAPI for role-based interviews, with voice interaction and automated feedback stored in SQLite.

AI Resume Screening Agent | Python, Flask, OpenAI, RAG

May 2025

Developed a resume-JD matching system using a RAG pipeline with ChromaDB, improving contextual accuracy by 28%.

PUBLICATION

Comprehensive Analysis of SPL Genes in Cucurbitaceae Species: Evolutionary Insights, Functional Diversity, and Regulatory Mechanisms — Iranian Journal of Science (April 2025)

[DOI: 10.1007/s40995-025-01818-w](https://doi.org/10.1007/s40995-025-01818-w)

EDUCATION

Centurion University of Technology and Management — Vizianagaram, Andhra Pradesh

2025

B.Tech, Computer Science and Engineering | CGPA: 8.55 / 10

CERTIFICATIONS

Infosys Springboard: Internship Completion Certificate (Internship 6.0)

Artificial Intelligence Primer Certification | Principles of Generative AI Certification (2025)

GenAI Academy — Google Cloud (2025) | Data Science Boot Camp (6 weeks) — GeeksforGeeks (2024)